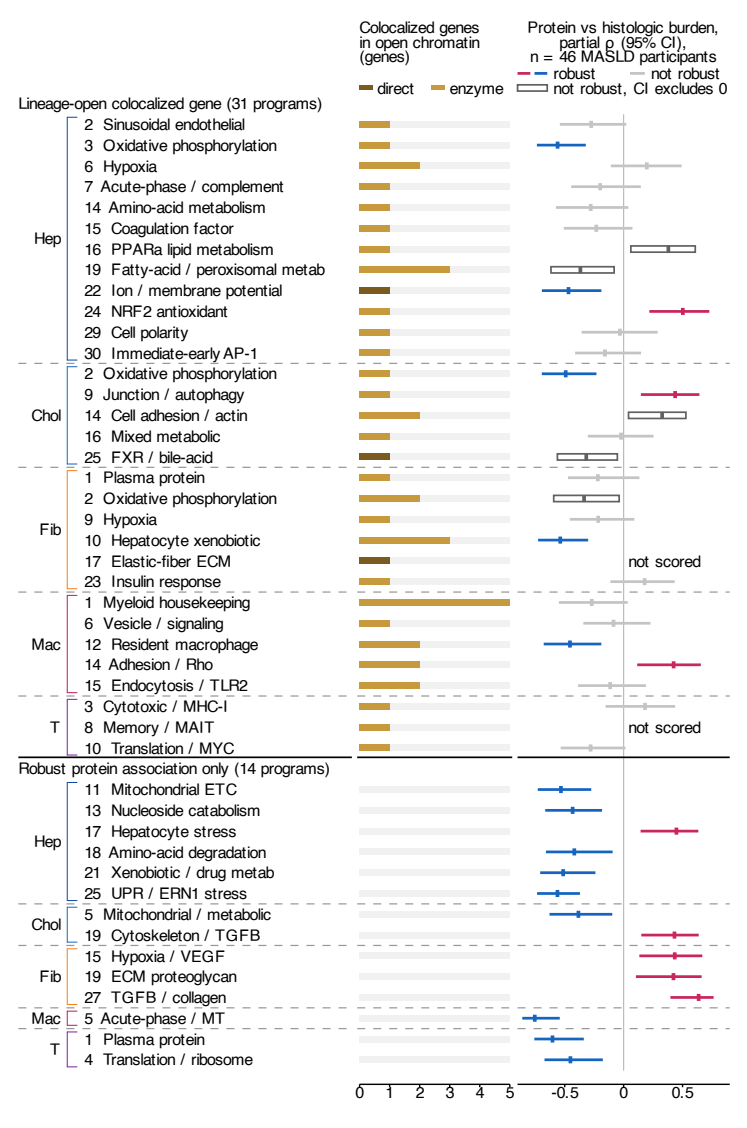
Colocalized genes in open chromatin (genes)
Protein vs histologic burden, partial ρ (95% CI), n = 46 MASLD participants
robust
not robust
direct
enzyme
not robust, CI excludes 0
Lineage-open colocalized gene (31 programs)
Hep
2 Sinusoidal endothelial
3 Oxidative phosphorylation
6 Hypoxia
7 Acute-phase / complement
14 Amino-acid metabolism
15 Coagulation factor
16 PPARa lipid metabolism
19 Fatty-acid / peroxisomal metab
22 Ion / membrane potential
24 NRF2 antioxidant
29 Cell polarity
30 Immediate-early AP-1
Chol
2 Oxidative phosphorylation
9 Junction / autophagy
14 Cell adhesion / actin
16 Mixed metabolic
25 FXR / bile-acid
Fib
1 Plasma protein
2 Oxidative phosphorylation
9 Hypoxia
10 Hepatocyte xenobiotic
17 Elastic-fiber ECM
23 Insulin response
not scored
Mac
1 Myeloid housekeeping
6 Vesicle / signaling
12 Resident macrophage
14 Adhesion / Rho
15 Endocytosis / TLR2
T
3 Cytotoxic / MHC-I
8 Memory / MAIT
10 Translation / MYC
not scored
Robust protein association only (14 programs)
Hep
11 Mitochondrial ETC
13 Nucleoside catabolism
17 Hepatocyte stress
18 Amino-acid degradation
21 Xenobiotic / drug metab
25 UPR / ERN1 stress
Chol
5 Mitochondrial / metabolic
19 Cytoskeleton / TGFB
Fib
15 Hypoxia / VEGF
19 ECM proteoglycan
27 TGFB / collagen
Mac
5 Acute-phase / MT
T
1 Plasma protein
4 Translation / ribosome
0 1 2 3 4 5
-0.5 0 0.5